

$\{x\} = x - n$. The graph of the function $y = \{x\}$ is shown in Fig. 27.

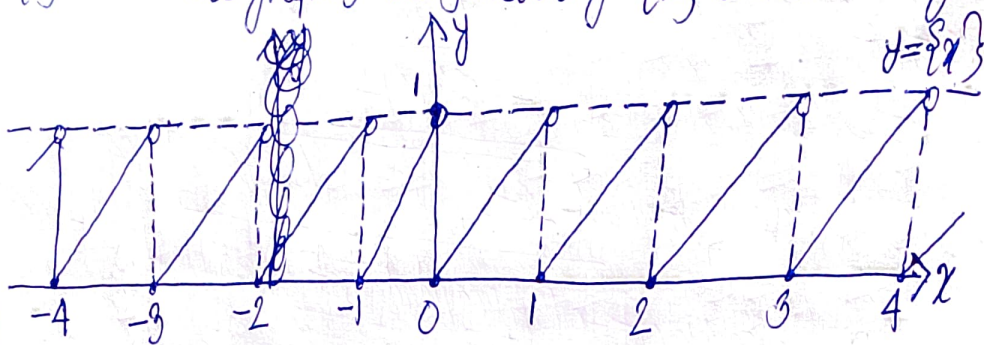


Fig 27

Take, for example, $x=1$. It is obvious that $\{x\}$ is defined at point $x=1$ ($\{1\}=0$). But does the function have the limit at $x=1$?

Reader: It clearly has no limit. In any δ -neighbourhood of point $x=1$ there may exist concurrently both the points at which $\{x\}$ assumes values greater than, for example, $\frac{2}{3}$, and the points at which $\{x\}$ assumes values less than $\frac{1}{3}$. It means that neither $b=1$ nor $b=0$ can be the limit of the function at point $x=1$, if only because it is impossible to find an adequate δ for $\varepsilon = \frac{1}{3}$.

Author: I see that you have come to be rather fluent in operating with limits of functions. My compliments. By the way, you have just proved the theorem on the uniqueness of the limit of function at a given point.

Theorem: A function cannot have two (or more) limits at a given point.

San: **Proof:** Let $f(x)$ be a given function, and suppose $b, c \in \mathbb{R}$ where $b \neq c$ be the limits of the function at x tending to a .
 $\therefore \forall \varepsilon > 0, \exists \delta$ s.t. $\forall x$ in the given domain, $|x-a| < \delta$, means $|f(x)-b| < \varepsilon$ and $|f(x)-c| < \varepsilon$. $\therefore$ $[a-\delta, a+\delta]$ and $[b-\varepsilon, b+\varepsilon]$ and $[c-\varepsilon, c+\varepsilon]$. Let $d = \min(b, c)$. $\therefore \forall x$ in $[a-\delta, a+\delta]$ and in the domain of f , $[d-\varepsilon, d+\varepsilon]$. We are taking the minimum because it is the intersection set. $\therefore |x-a| < \delta$, then $|f(x)-d| < \varepsilon$.

So, d is the unique limit of $f(x)$ at x tending to a . $\square$